CHAPTER

25

Special Imaging Techniques

This chapter presents four specific aspects of image processing. First, ways to characterize the *spatial resolution* are discussed. This describes the minimum size an object must be to be seen in an image. Second, the *signal-to-noise ratio* is examined, explaining how faint an object can be and still be detected. Third, *morphological* techniques are introduced. These are nonlinear operations used to manipulate binary images (where each pixel is either black or white). Fourth, the remarkable technique of *computed tomography* is described. This has revolutionized medical diagnosis by providing detailed images of the interior of the human body.

Spatial Resolution

Suppose we want to compare two imaging systems, with the goal of determining which has the best spatial resolution. In other words, we want to know which system can detect the smallest object. To simplify things, we would like the answer to be a *single number* for each system. This allows a direct comparison upon which to base design decisions. Unfortunately, a single parameter is not always sufficient to characterize all the subtle aspects of imaging. This is complicated by the fact that spatial resolution is limited by two distinct but interrelated effects: *sample spacing* and *sampling aperture size*. This section contains two main topics: (1) how a single parameter can best be used to characterize spatial resolution, and (2) the relationship between sample spacing and sampling aperture size.

Figure 25-1a shows profiles from three circularly symmetric PSFs: the pillbox, the Gaussian, and the exponential. These are representative of the PSFs commonly found in imaging systems. As described in the last chapter, the pillbox can result from an improperly focused lens system. Likewise, the Gaussian is formed when random errors are combined, such as viewing stars through a turbulent atmosphere. An exponential PSF is generated when electrons or x-rays strike a phosphor layer and are converted into